NuraMed: Product Requirements Document

HT-OPEN | Open Innovation, HealthTech & Accessibility | Prototype v1.0 | Hackathon build

1. Overview

Product: NuraMed, a bridge between patients and doctors for people who cannot travel back and forth for routine follow-ups and availibilty of doctors.

One-line pitch: Reach your doctor without the journey, by app, by chat, or by a simple phone call.

Problem: Patients across the country travel long distances only to get a dosage reviewed, a follow-up checked, or to learn whether a doctor is available. Elderly, low-literacy and feature-phone users are left out of app-only digital health tools.

Solution: One shared backend powering a patient web app (PWA), four core portals, an AI health assistant, a toll-free IVR line, and two side portals (Admin, Health Worker/Pharmacy).

2. Goals and Non-Goals

Goals

- Let patients request prescription and dosage reviews remotely, with the doctor making every decision.
- Show which doctors are available, with emergency guidance always first.
- Enable chat, audio and video consults with a low-bandwidth mode.
- Serve non-smartphone users through a toll-free IVR with callback.
- Give basic first-aid and self-care guidance through a safe, source-grounded AI assistant which can track all the things of the patient .
- Demonstrate improved access, clarity and task completion without overstating clinical outcomes.

Non-Goals

- No diagnosis by the system or the AI.
- No automatic dosage change, and no AI recommendation of medicines or doses.
- No replacement for emergency services (108/112).
- No real patient data, and no claims of clinical outcomes or accuracy percentages.

3. Users and Personas (all fictional)

Persona	Profile	Primary need
Ramesh, 62	Smartphone user, on BP medication	Ask for dosage review without a clinic visit
Kamla Devi, 70	Feature phone only, Hindi speaker	Reach a doctor by calling a number
Dr. Meera	General physician	Review requests quickly with full context and an audit trail
Sunita	ASHA worker	Raise requests on behalf of patients who cannot do it themselves
Admin	Program coordinator	Manage doctors, monitor requests and call logs
Pharmacist	Local pharmacy	View only approved prescriptions

4. System Architecture (high level)

- **Patient App (PWA):** hosts Portals 1, 2, 3 and the AI assistant.
- **IVR (Twilio Voice):** Portal 4, a phone interface to the same backend work same as all the features in my app using toll free no. like dial 1 to track patitent app similarly (like jio call free no . u can add the our system architech).
- **Doctor Dashboard:** receives requests from the app, IVR and health workers in one queue.
- **Shared Backend + DB:** single source of truth, so an IVR request appears in the same doctor queue as an app request.
- **Side Portals:** Admin, and Health Worker/Pharmacy.
- **Suggested stack:** React/Next.js, Node or FastAPI, three .js or 3d , Twilio Voice and SMS, WebRTC or Jitsi, Claude API for the assistant.

5. Global Requirements

ID	Requirement
G-1	Every screen shows an emergency banner: "In an emergency call 108 / 112 immediately."

ID	Requirement
G-2	Role-based access: patient, doctor, health worker, pharmacy, admin.
G-3	Every clinical action (request, approve, modify, reject) is written to an immutable audit log with user, time and action.
G-4	UI in Hindi and English; Rajasthani listed as a roadmap language.
G-5	Accessibility: large-font toggle, high contrast, text-to-speech, simple icon-led navigation.
G-6	Only fictional or de-identified data in the prototype.
G-7	Consent checkbox before any data or photo submission.

6. Portal 1: Prescription Review

Purpose: Let a patient request a medicine or dosage review remotely. The doctor decides.

Core (Must)

ID	Requirement
P1-1	Patient submits a review request: current medicine (from their existing prescription), issue description, and reason (side effect, dose too high or low, missed dose, other).
P1-2	Doctor sees a request queue and can **Approve, Modify or Reject**, with mandatory notes on modify and reject.
P1-3	The system never changes a prescription without doctor action. Request status is Pending until reviewed.
P1-4	Patient is notified of the outcome, with the reason on rejection.

ID	Requirement
P1-5	Full audit trail per request: submitted, viewed, decision, notes, timestamps.

Add-ons

ID	Feature	Priority
P1-6	Side-effect symptom checklist form (dizziness, nausea, rash, etc.) attached to the request	Should
P1-7	Prescription history timeline	Should
P1-8	Doctor "Urgent" flag, auto-suggested when red-flag symptoms are ticked	Should
P1-9	Approved prescription delivered as PDF and SMS	Should
P1-10	Medicine reminders by SMS/WhatsApp (simulated in prototype)	Could
P1-11	Doctor can call the patient from the request card	Could

Acceptance criteria

- Dosage-change request leaves the stored dosage unchanged until a doctor approves.
- Rejected request shows the patient a reason.
- Each decision appears in the audit log.

7. Portal 2: Doctor Availability

Purpose: Show which doctors are reachable now, especially when a patient needs help quickly.

Core (Must)

ID	Requirement
P2-1	Emergency banner pinned at the top: 108 / 112, tap to dial.
P2-2	Doctor list with status (Online, Busy, Offline) updated by the doctor.

ID	Requirement
P2-3	Filter by specialty.
P2-4	Selecting an available doctor leads into Portal 3. Offline doctors show alternates.

Add-ons

ID	Feature	Priority
P2-5	Language filter (Hindi, Rajasthani, English)	Should
P2-6	"Next available slot" display	Should
P2-7	On-call roster view	Should
P2-8	Nearest hospital / PHC list (static fictional data)	Should
P2-9	Nearby doctors map	Could

Acceptance criteria

- Emergency banner is visible without scrolling on every screen.
- Request to an offline doctor returns "Doctor unavailable" with a suggested alternate.

8. Portal 3: Doctor-Patient Connect

Purpose: Real-time or asynchronous communication between patient and doctor.

Core (Must)

ID	Requirement
P3-1	Text chat between patient and doctor.
P3-2	Audio call; video call as a stretch goal (WebRTC/Jitsi).
P3-3	Low-bandwidth mode: auto-suggest audio-only or text when connection is weak.
P3-4	Callback request button when a doctor is busy.

Add-ons

ID	Feature	Priority
P3-5	Pre-call form (symptoms, duration, current medicines) shared with the doctor	Should
P3-6	Appointment booking	Should
P3-7	Post-consult summary note visible to the patient	Should
P3-8	Accessibility: text-to-speech of doctor messages, large font	Should
P3-9	Chat summary and photo attachment passed in from the AI assistant	Should

Acceptance criteria

- On weak connectivity the user is offered audio-only or text.
- Pre-call form content appears in the doctor's view before the session starts.

9. Portal 4: Toll-Free IVR (Twilio)

Purpose: Give feature-phone, elderly and low-literacy users the same access through a phone call.

Core (Must)

ID	Requirement
P4-1	Hindi voice menu: **1** = talk to a doctor, **2** = prescription review request, **3** = doctor availability, **0** = human callback.
P4-2	Option 2 records the patient's spoken problem and creates a request in the shared doctor queue.
P4-3	Callback model: the doctor calls the patient back.
P4-4	Wrong key press repeats the menu; pressing 0 routes to callback.

Add-ons

ID	Feature	Priority
P4-5	SMS confirmation after the call	Should
P4-6	Missed-call-back service (patient gives a missed call, system calls back at no cost to the patient)	Should
P4-7	Voice message transcription shown on the doctor's dashboard	Could
P4-8	Regional language option (Rajasthani)	Could
P4-9	Spoken red-flag guidance: if the caller says chest pain or breathlessness, the IVR immediately advises calling 108	Could

Prototype note

If a live Twilio number is not available, provide an **IVR simulator** in the web UI that follows the identical call flow and writes to the same queue.

Acceptance criteria

- An IVR request appears in the doctor dashboard in the same format as an app request.
- Invalid input never dead-ends the caller.

10. AI Health Assistant ("Setu Sahayak")

Purpose: Basic guidance on minor health issues and first aid, grounded in verified sources, with clear escalation to a doctor.

Approach

Retrieval-augmented generation (RAG) over 10 to 20 verified documents (WHO, MoHFW, Red Cross first-aid guides), a strict system prompt, and the Claude API. No model training or fine-tuning.

Requirements

ID	Requirement	Priority
AI-1	Text chat in Hindi, English and Hinglish for minor issues: fever, cold and cough, stomach upset, minor cuts and burns, dehydration, heat protection	Must
AI-2	Red-flag detection (chest pain, breathlessness, heavy bleeding, etc.) shows the 108/112 warning immediately and pauses normal chat	Must
AI-3	Hard rules: no diagnosis, no medicine names or doses, no replacement for doctor advice; redirects to Portal 1 or a doctor	Must
AI-4	"Connect to a doctor" button at the end of the conversation leading to Portal 3, with chat summary	Must
AI-5	Disclaimer under every answer: "This is general information, not medical advice."	Must
AI-6	Wound photo upload with consent: AI describes only what is visible, gives general first aid, never gives a verdict	Should
AI-7	Unclear photos prompt a retake; photos with serious signs prompt "see a doctor / call 108"	Should
AI-8	"Send to doctor" attaches the photo and summary to Portal 1 or 3	Should
AI-9	Photo privacy: strip EXIF and location data; fictional or stock images only in the demo	Must
AI-10	Out-of-scope questions are politely declined	Should

11. Side Portal A: Admin

ID	Requirement	Priority
AD-1	Add, edit and deactivate doctors; set specialty and languages	Must

ID	Requirement	Priority
AD-2	Dashboard: requests received, completed, pending, and source (app, IVR, health worker)	Must
AD-3	Call logs and IVR usage	Should
AD-4	Average time from request to doctor response (supports the expected-benefit demo)	Should
AD-5	Audit log viewer	Should

12. Side Portal B: Health Worker (ASHA) and Pharmacy

Health Worker / Family Member

ID	Requirement	Priority
HW-1	Submit a prescription review request or doctor connect on behalf of a patient, with the patient's consent recorded	Must
HW-2	Use the AI assistant on the patient's behalf	Should
HW-3	View the status of requests the worker submitted	Should

Pharmacy

ID	Requirement	Priority
PH-1	View **approved** prescriptions only (read-only)	Must
PH-2	Mark a prescription as fulfilled	Could

13. Demo User Journeys

- **Journey A (app):** Ramesh opens the app, submits a dosage review with a side-effect checklist, Dr. Meera approves with a note, Ramesh receives the updated prescription as PDF and SMS.
- **Journey B (IVR):** Kamla Devi dials the toll-free number, presses 2, speaks her problem, the request reaches the doctor queue, and the doctor calls back.

- **Journey C (emergency):** A patient opens availability, the 108/112 banner appears first, then nearby available doctors are shown.
- **Journey D (AI assistant):** A patient uploads a fictional wound photo, receives first-aid steps, is advised to see a doctor, and forwards the photo to the doctor.
- **Journey E (health worker):** Sunita raises a request for a patient without a phone, and the pharmacy later sees the approved prescription.

14. Representative Test Cases

#	Test	Expected result
1	Patient requests a higher dose	Request goes to the doctor; dosage unchanged until approved
2	Doctor rejects a request	Patient notified with reason
3	Request sent to an offline doctor	"Doctor unavailable" plus alternate suggested
4	Wrong key pressed on IVR	Menu repeats; 0 gives human callback
5	Patient types "chest pain, can't breathe"	Immediate 108/112 warning
6	Weak internet	Audio-only or SMS fallback offered
7	"Which medicine should I take, what dose?" in chatbot	Refuses and redirects to Portal 1 or a doctor
8	Deep wound photo	First aid plus urgent doctor advice
9	Blurry photo	Asks for a clearer photo
10	"Is this cancer?"	No diagnosis; advises seeing a doctor
11	Pharmacy opens a pending prescription	Not visible; only approved ones shown
12	IVR request	Appears in doctor queue with source "IVR"

15. Privacy, Safety and Accessibility

- Fictional or de-identified data only; no real patient records.
- Doctors make all clinical decisions; the system handles communication and workflow.
- Not an emergency service; emergency guidance is always visible.
- Red-flag detection is keyword-based and is not diagnostic.

- Consent is captured before data or photo submission; photo metadata is stripped.
- Role-based access and audit logging throughout.
- Production use would require DPDP Act compliance, encryption, telemedicine guideline adherence, and clinical validation.

16. Expected Benefit (carefully worded)

SetuCare may reduce unnecessary repeat travel for routine follow-ups, speed up follow-up requests, and let non-smartphone users reach a doctor. This prototype demonstrates improved access and task completion only. It makes no claims about health outcomes or AI accuracy.

17. Limitations

- AI responses can be wrong; photo assessment is limited by lighting, skin tone and camera quality.
- IVR is menu-based only; there is no voice AI.
- Reminders, WhatsApp and map features are simulated or static in the prototype.
- Availability data is fictional, not live hospital data.
- No clinical validation has been performed.

18. Scope and Priorities

Priority	Scope
Must demo	Portal 1 approval flow with audit trail, Portal 2 emergency banner and list, Portal 4 IVR (live or simulated) with callback, doctor dashboard, AI chatbot with red-flag escalation
Should demo	Portal 3 chat and audio, wound photo upload and forward to doctor, symptom checklist, PDF/SMS prescription, Admin dashboard
Slide only	Video, map, WhatsApp reminders, Rajasthani IVR, transcription, voice AI, pharmacy fulfilment

19. Success Metrics for the Demo

These are prototype indicators, not clinical outcomes.

- Request-to-decision flow works end to end in all five journeys.
- IVR and app requests land in one doctor queue.
- All 12 test cases pass.
- Red-flag input always triggers the emergency warning.
- A user can complete a prescription request in under three minutes in the demo.

20. Suggested Build Order

1. Shared data model, roles and audit log
2. Portal 1 and doctor dashboard
3. Portal 2 with emergency banner
4. IVR simulator, then Twilio if time allows
5. AI assistant with red-flag rules, then photo upload
6. Portal 3 chat and audio
7. Admin and Health Worker/Pharmacy views
8. Demo script, test cases and limitations slide